

Benchmarking Variational Quantum Classifiers Against Classical SVCS with US Unemployment Rate

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Introduction

Describe your motivation, initial idea that you worked on based on the project proposal to investigate, and if applicable any prior work, interest in the topic.

Answer. We want to investigate where the advantages of current quantum models lie, and where they may fall short against classical models, through the application of realistic sets of data. Since most real-world data is inherently non-linear, the data we used is aimed to create a complex setting that requires models to capture intricate patterns and variances. The current consensus is that while quantum models offer a novel approach to data processing, traditional models have maintained a higher degree of accuracy and stability, and this difference serves as the main point of our investigation. A major goal of this project is to understand the architectural gaps that prevent emerging quantum models from surpassing the proven benchmarks of classical computation in its current state.

The primary motive for incorporating data on unemployment rates in the US is that a variety of factors may cause them to fluctuate. With nearly a century of historical data to potentially cover, including periods of major financial crisis, it is one of the key statistics to represent the economy's overall health. Without historical context, including background knowledge of a time period's wars and global events, the fluctuation of this data may be considered near impossible to predict. In our decision to limit the selection to this decade, our group incorporates the shock of COVID-19 and its aftermath in both types of models.

Data Description

Describe what classical dataset you used and what quantum encoding method you used in your project. Make sure to clearly explain determined which datasets you used in this project, the criteria, source, data and information types in detail, associated documentation and any other supporting materials.

Answer. In this project, the primary dataset for our investigation was from the Federal Reserve Economic Data (FRED), which is maintained by the Federal Reserve Bank of St. Louis. Specifically, we utilized the unemployment rates that were recorded as a practical application test. This time-series data is characterized by monthly fluctuations and erratic economic cycles; it represents the percentage of the labor force that is unemployed, as collected by the U.S. Bureau of Labor Statistics. The time-frame for this study covers January 1, 2020 to January 1, 2026, capturing the significant shocks of the early 2020s and the subsequent economic stabilization that followed.

Since the raw data consists of continuous percentages, we cleaned the dataset in order to remove all of the missing or “NULL” data points. This was accomplished using forward filling to maintain chronological integrity and avoid gaps in the time-series sequence. To capture short-term trends, we transformed the raw percentages into a binary classification task by using a rolling 4-month window. The model uses the previous four months of data as features to predict a binary target—whether the unemployment rate will rise (value of 1) or fall (value of 0) in the subsequent month.

We developed two different types of predictive architectures: a Classical Support Vector Machine (SVM) and a Hybrid Quantum Classical Classifier (VQC). The classical model we developed uses a standard SVM using a Radial Basis Function (RBF) kernel, which is a proven benchmark for high-dimensional classification. With the quantum model, we implemented a quantum kernel estimator using the ZZ feature map to encode our classical features into a high-dimensional Hilbert Space using quantum entanglement. To ensure the models could generalize to unseen data without underfitting or overfitting, we utilized a 70/30 stratified train-test split. This results in a total of 22,785 samples, with 15,340 dedicated to training and 6,836 dedicated to testing. This sample size was necessary to provide enough iterations for the COBYLA optimizer to attempt to converge during the quantum circuit training phase.

Quantum Model Description Analysis

Discuss what specific quantum computing, QML model you used (i.e QSVC, VQC) and explain how you used it following the QML project workflow tutorial provided to you during the class.

Answer. We implemented a hybrid quantum-classical model, specifically a Variational Quantum Classifier (VQC), using a quantum kernel approach for this project. This aligns with the current practical approach in Quantum Machine Learning, where quantum circuits are primarily used for feature transformation rather than full model optimization.

In following the QML workflow introduced in class, several structured steps outlined our implementation. First, after preparing and splitting the data, we performed quantum feature encoding using the ZZ Feature Map. This step is critical because it maps classical input data into a quantum Hilbert space, where relationships between points can become more separable. Unlike classical kernels, the quantum feature map uses entanglement between qubits to capture correlations between features. For FRED's unemployment dataset, 4 qubits represented the 4-month feature windows we chose. This allowed us to apply a second-order expansion that uses entanglement (CNOT gates) to capture correlations between monthly lags that classical kernels might overlook. Following this, we constructed a quantum kernel matrix, which measures similarity between all pairs of data points in the transformed quantum space. Conceptually, this step allows us to evaluate how "close" two data points are following quantum encoding, which is an essential step for classification.

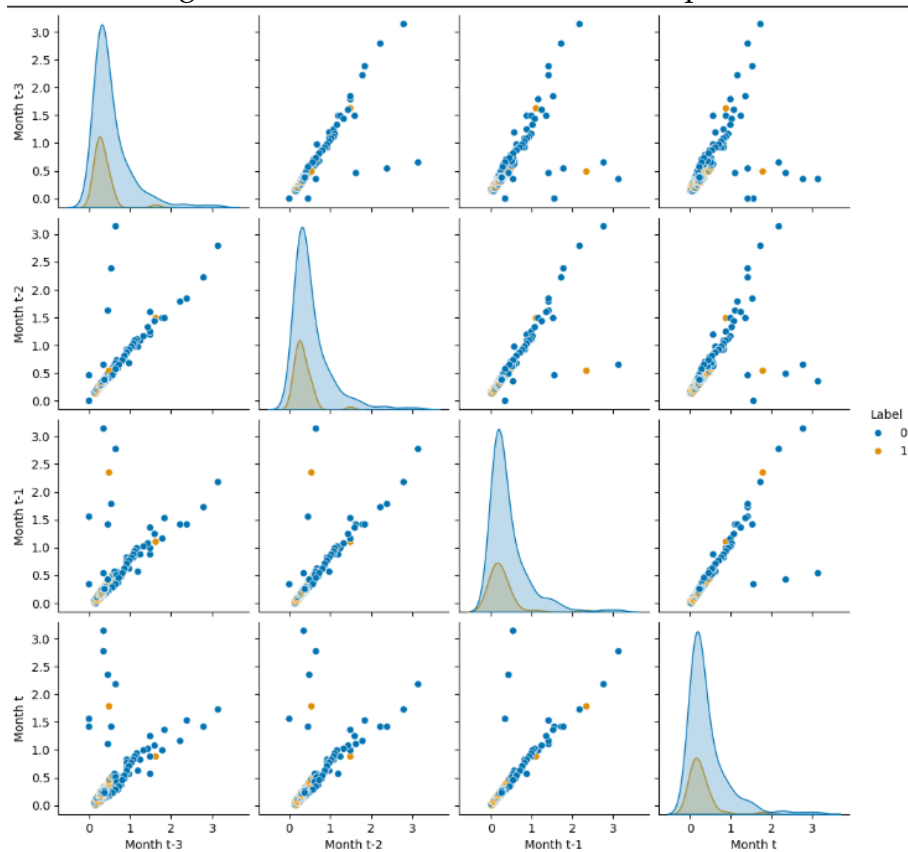
Because quantum computers are not yet efficient at solving optimization problems, we followed a hybrid approach (VQC) by using a classical SVM trained on the precomputed quantum kernel. This allowed us to still leverage strong classical optimization while incorporating quantum feature transformations. This design choice is consistent with current limitations in quantum hardware and is widely used in QML research.

In addition to the kernel-based approach, our implementation also explored a parameterized quantum circuit (Ansatz) to simulate learning behavior. This circuit was combined with the feature map and optimized using a classical optimizer, allowing us to approximate a variational classifier. However, due to computational constraints, this component was kept relatively shallow. During training, we executed quantum circuits multiple times to estimate probabilities. The outputs were interpreted using a parity-based measurement, converting quantum states into binary classification predictions. The model's performance was evaluated using accuracy on a held-out test set. Overall, the implementation of this hybrid VQC framework served as a robust diagnostic tool for assessing the current state of quantum technologies in economic forecasting. While the model successfully navigated the complex QML workflow— from ZZ feature entanglement to COBYLA optimization – it ultimately highlighted the performance gap that persists between theoretical quantum potentials and practical applications.

Model Development and Application

What types of models you used to describe the data (regression, classification, , etc.), patterns/ trends you found, visual approaches that helped you choose models, and or variables (type/ number) in the model, other parameter choices or settings for the models (e.g. number of iterations for VQC model, quantum kernels). Discuss what specific quantum computing, QML model you used (i.e QSVS, VQC). Explain how you used it following the QML project workflow tutorial provided to you during the class earlier. Apply the models to assess model performance (i.e. predict). Discuss the confidence in your results.

Figure 1: Results across a four-month period



Answer. In this project, we developed and conducted a comprehensive benchmark of several classification architectures: five distinct classical models and a hybrid Variational Quantum Classifier (VQC). The classical suite included Logistic Regression, Support Vector Classifier (SVC), K-Nearest Neighbors (KNN), Random Forest, and Gradient Boosting. These were compared against a VQC framework that utilized two separate quantum encoding strategies: the ZZ Feature Map and the Pauli Feature Map. Both sets of models were applied to a realistic dataset consisting of transformed civilian unemployment rates, where the objective was to perform a binary classification to predict whether unemployment would increase or decrease in the following month. The unemployment dataset required additional preprocessing, where we created 4-month sliding windows so that the model could learn short-term trends and predict

future changes. This transformation allowed us to treat time-series data as a classification problem. For the classical model, we primarily focused on the SVM with an RBF kernel, which is effective for identifying complex decision boundaries in datasets where classes are not linearly separable. For the quantum model, we utilized 4 qubits, which match our feature count and maps data into a high-dimensional Hilbert Space. The VQC models employed a Two-Local (RyCNOT) Ansatz circuit and were optimized using the COBYLA algorithm with a budget of up to 100 maximum iterations. Both model categories were trained on a 70% stratified training set and evaluated on a 30% test set. The results indicated that the classical models, particularly the Random Forest and KNN, achieved higher and more stable accuracies (79%). While the quantum models achieved similar raw test accuracies (78–79%), they exhibited a significant "accuracy paradox". Loss tracking for the VQC showed gradual improvement and convergence, yet the models primarily learned to predict the majority class, resulting in near-zero recall for the minority class.

Our confidence in the results is moderate. While the workflow was consistent, limitations such as small dataset size, limited iterations, and quantum simulation constraints likely prevented the quantum model from reaching optimal performance. The figure above shows a pair plot of unemployment features. The display includes the distribution and pairwise relationships for the 4-month sliding window features. The high degree of overlap and linear correlation between lags ($t-3$ to t) justifies the use of kernels, such as RBF and ZZ Feature Map, to attempt to distinguish between the classification labels.

Conclusions and Discussion

Describe your conclusions; interpret the results, predictions you made, the models and their characteristics, and give a summary of what changed as you went through the project (QML model choices, etc.), what you would do next, or do differently in a subsequent exploration.

Answer. From our results, we conclude that classical machine learning models currently outperform quantum models, but not by a large margin. While they are outclassed in raw metrics, the VQC model demonstrated that it could produce better generalization than classical models. Both methods had nearly equal accuracies between training and testing sets according to the results. The classical SVM achieved higher accuracy and required significantly less computation, while the quantum model showed potential but was limited in performance due to quantum hardware limitations. These results suggest that the main limitation of QML is not the concept itself, but the current state of technology. Quantum models are restricted by factors such as limited qubits, shallow circuits, and expensive simulations, which prevent them from their theoretical advantages.

During the project, our approach evolved as we adjusted for these constraints. We reduced dataset size, controlled the complexity, restricted iterations, standardized depth, and relied on hybrid methods, which showed us how dependent QML currently is on classical components. This was done through testing predictions through both the ZZ and Pauli feature maps and comparing their results against classical models such as K-Nearest Neighbor (KNN), Random Forest, Regular Regression, and XGBoost. The quantum models using these feature maps were able to achieve relatively high test accuracies (78-79%), which weren't far off of the accuracies from the classical methods. The only issue with the quantum models was that it was able to primarily predict the majority class (higher unemployment rates) rather than the actual direction of the unemployment rate. The ZZ Feature Map settled into its loss function earlier than the Pauli Feature Map and also maintained higher accuracy, but this was attributed to it focusing almost exclusively on the majority class. On the other hand, the Pauli Feature map took longer to settle (converge) but showed signs of "taking risks" to learn and identify the minority class more effectively than the ZZ Feature map.

In future work, we would test different realistic datasets, increase training iterations, explore models like QSVC or quantum neural networks, and run experiments on real quantum hardware. These experiments might yield new and interesting findings, like what was found in the differences between the feature map behavior. Classical models remain more practical today, but QML is promising and quantum approaches may become competitive in the near future as technology improves.